

CLUSTER SETUP

```
default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your c
cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-09-15 08:09:03 [i] using Kubernetes version 1.34
2026-09-15 08:09:03 [i] creating EKS cluster "e-commerce" in "us-east-1" region with
2026-09-15 08:09:03 [i] if you encounter any issues, check CloudFormation console or try 'eks
ctl utils describe-stacks --region=us-east-1 --cluster=e-commerce'
2026-09-15 08:09:03 [i] Kubernetes API endpoint access will use default of {publicAccess=true
, privateAccess=false} for cluster "e-commerce" in "us-east-1"
2026-09-15 08:09:03 [i] CloudWatch logging will not be enabled for cluster "e-commerce" in "
us-east-1"
2026-09-15 08:09:03 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-
types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=e-commerce'
2026-09-15 08:09:03 [i] default add-ons vpc-cni, kube-proxy, coredns, metrics-server were not
specified, will install them as EKS add-ons
2026-09-15 08:09:03 [i]
2 sequential tasks: { create cluster control plane "e-commerce",
  2 sequential sub-tasks: {
    1 task: { create add-ons },
    wait for control plane to become ready,
  }
}
2026-09-15 08:09:03 [i] building cluster stack "eksctl-e-commerce-cluster"
2026-09-15 08:09:03 [i] deploying stack "eksctl-e-commerce-cluster"
2026-09-15 08:09:33 [i] waiting for CloudFormation stack "eksctl-e-commerce-cluster"
2026-09-15 08:10:03 [i] waiting for CloudFormation stack "eksctl-e-commerce-cluster"
2026-09-15 08:11:03 [i] waiting for CloudFormation stack "eksctl-e-commerce-cluster"
```

```
root@ip-172-31-30-22~
[root@ip-172-31-30-22 ~]# aws eks update-kubeconfig --name e-commerce --region us-east-1
Added new context arn:aws:eks:us-east-1:044771288944:cluster/e-commerce to /root/.kube/config
[root@ip-172-31-30-22 ~]# kubectl get no
NAME                                STATUS    ROLES    AGE     VERSION
ip-192-168-10-181.ec2.internal      Ready    <none>   7m53s   v1.34.10-eks-cb19647
ip-192-168-57-64.ec2.internal       Ready    <none>   7m49s   v1.34.10-eks-cb19647
ip-192-168-60-114.ec2.internal      Ready    <none>   7m52s   v1.34.10-eks-cb19647
[root@ip-172-31-30-22 ~]#
```

AWS CONSOLE AFTER THE SET-UP

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:v=3;\$case=tags:true%5C,client:false;\$regex=tags:false%5C,client:false

EC2 > Instances

Instances (4) Info

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4
e-commerce-cluster-nodes-Node	i-063bce8336b0be0bb	Running	c7i-flex.large	3/3 checks passed	us-east-1a	ec2-3-92-81
e-commerce-cluster-nodes-Node	i-0ca19d47c3394cd56	Running	c7i-flex.large	3/3 checks passed	us-east-1b	ec2-184-72
e-commerce-cluster-nodes-Node	i-09954add12f3d5b57	Running	c7i-flex.large	3/3 checks passed	us-east-1b	ec2-3-87-22
ECOM_MICROSERVICES	i-0e31b16ddc28a3fa1	Running	m7i-flex.large	3/3 checks passed	us-east-1c	ec2-52-91-1

All instances

Instance alarm states

In alarm

None Show instances in alarm

Service health

EC2 service status

Operating normally

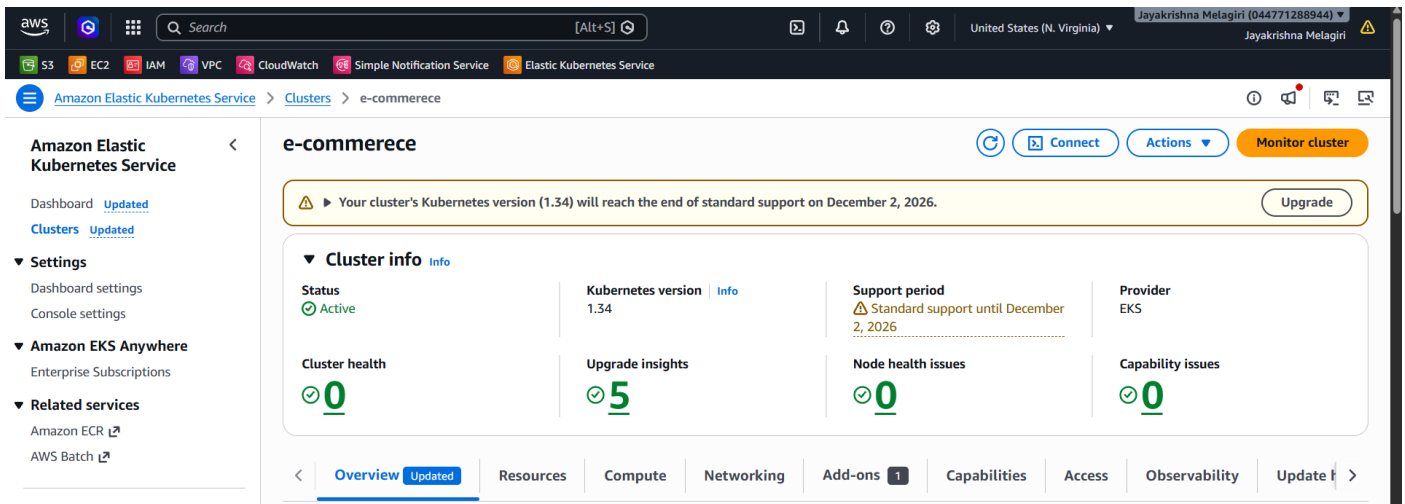
Health events affecting EC2 instances

Open events

None Show instances

Upcoming events

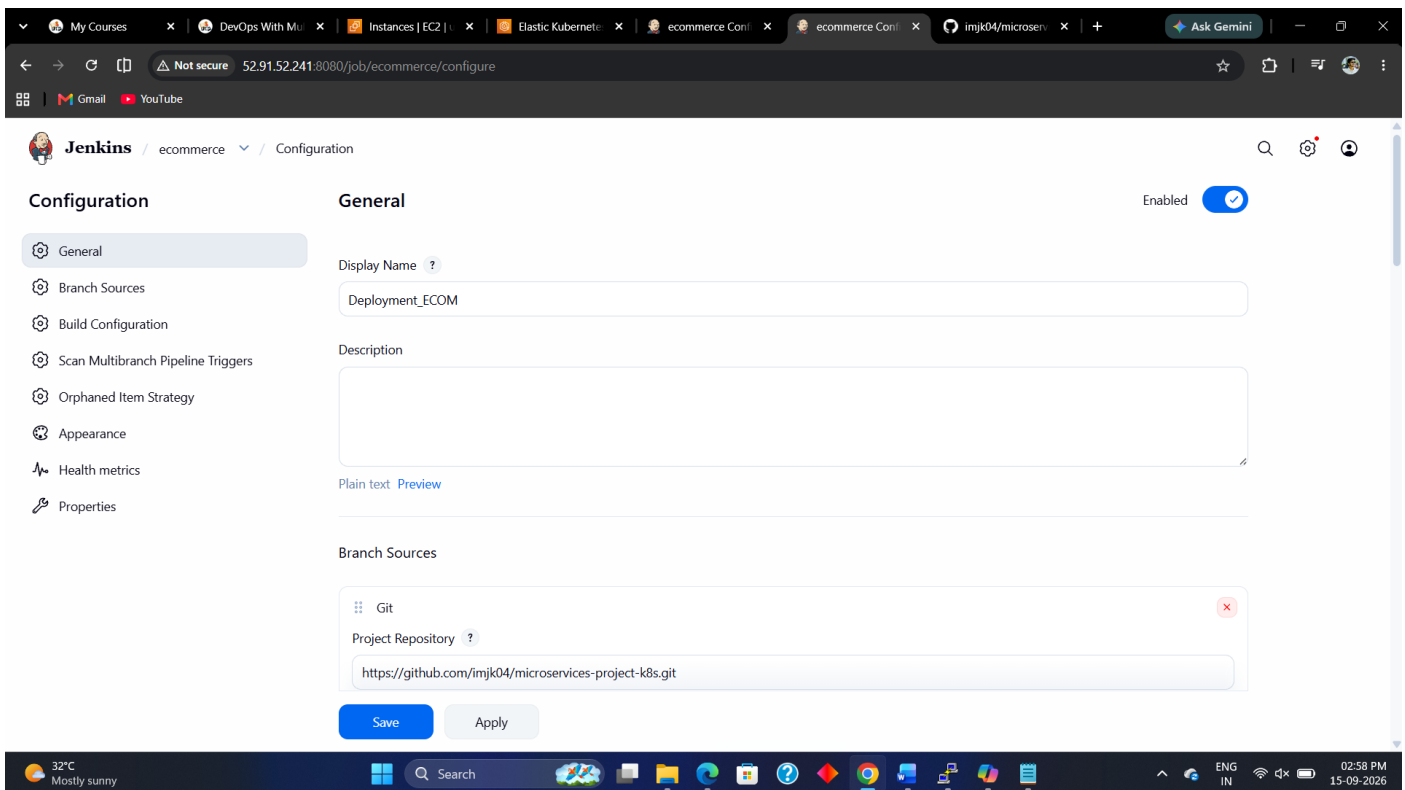
None Show instances



CREATING THE SERVICE ACCOUNT TO ENSURE JENKINS CREATE THE K8S RESOURCE

```
[root@ip-172-31-30-22 ~]# vim role.yml
[root@ip-172-31-30-22 ~]# kubectl create -f role.yml -n ecom
role.rbac.authorization.k8s.io/deploy-role created
[root@ip-172-31-30-22 ~]# vim service_account.yml
[root@ip-172-31-30-22 ~]# kubectl create -f service_account.yml -n ecom
serviceaccount/jenkins created
[root@ip-172-31-30-22 ~]# vim rolebinding.yml
[root@ip-172-31-30-22 ~]# kubectl create -f rolebinding.yml -n ecom
rolebinding.rbac.authorization.k8s.io/deploy-rolebinding created
[root@ip-172-31-30-22 ~]# vim secrets.yml
[root@ip-172-31-30-22 ~]# kubectl create -f secrets.yml -n ecom
secret/mysecretname created
[root@ip-172-31-30-22 ~]#
```

MULTI BRANCH PIPELINE SETUP



Jenkins / Deployment_ECOM

Status

Configure

Scan Multibranch Pipeline Now

Scan Multibranch Pipeline Log

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

Project Relationship

Check File Fingerprint

Rename

Pipeline Syntax

Credentials

Build Queue

Build Executor Status

Folder name: ecommerce

Branches (12)

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	adservice	46 min #1	N/A	4 min 58 sec
✓	☀	cartservice	46 min #1	N/A	3 min 20 sec
✓	☁	checkoutservice	35 min #2	46 min #1	39 sec
✓	☀	currencyservice	46 min #1	N/A	3 min 6 sec
✓	☁	deployment-(CD)	47 sec #4	22 min #3	3.1 sec
✓	☁	emailservice	26 min #3	35 min #2	45 sec
✓	☀	frontend	46 min #1	N/A	3 min 17 sec
✓	☀	loadgenerator	46 min #1	N/A	3 min 31 sec
✓	☀	paymentservice	46 min #1	N/A	4 min 35 sec

Jenkins / Deployment_ECOM

Rename

Pipeline Syntax

Credentials

Build Queue

Build Executor Status

(0 of 5 executors busy)

✓	☁	deployment-(CD)	47 sec #4	22 min #3	3.1 sec
✓	☁	emailservice	26 min #3	35 min #2	45 sec
✓	☀	frontend	46 min #1	N/A	3 min 17 sec
✓	☀	loadgenerator	46 min #1	N/A	3 min 31 sec
✓	☀	paymentservice	46 min #1	N/A	4 min 35 sec
✓	☀	productcatalogservice	46 min #1	N/A	4 min 33 sec
✓	☁	recommendationservice	26 min #2	46 min #1	43 sec
✓	☀	shippingservice	46 min #1	N/A	2 min 43 sec

ACCESSING THE APPLICATION

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP
		PORT (S)	AGE
adservice	ClusterIP	10.100.238.146	<none>
		9555/TCP	2m21s
cartservice	ClusterIP	10.100.145.101	<none>
		7070/TCP	2m21s
checkoutservice	ClusterIP	10.100.174.71	<none>
		5050/TCP	2m22s
currencyservice	ClusterIP	10.100.232.182	<none>
		7000/TCP	2m21s
emailservice	ClusterIP	10.100.239.32	<none>
		5000/TCP	2m22s
frontend	NodePort	10.100.36.191	<none>
		80:30287/TCP	2m22s
frontend-external	LoadBalancer	10.100.121.103	a4f10a94baa6d49b0afa6c60766e7cc4-11646
63195.us-east-1.elb.amazonaws.com		80:30945/TCP	2m22s
paymentservice	ClusterIP	10.100.115.188	<none>
		50051/TCP	2m21s
productcatalogservice	ClusterIP	10.100.12.89	<none>
		3550/TCP	2m21s
recommendationservice	ClusterIP	10.100.253.19	<none>
		8080/TCP	2m22s
redis-cart	ClusterIP	10.100.224.47	<none>
		6379/TCP	2m21s
shippingservice	ClusterIP	10.100.122.253	<none>
		50051/TCP	2m21s

```
[root@ip-172-31-30-22 ~]# kubectl get po -n ecom
NAME                                READY    STATUS    RESTARTS    AGE
adservice-786959b787-xd5xr         1/1     Running   0           10m
cartservice-6dd7c5b4cb-gdd71      1/1     Running   0           10m
checkoutservice-5bdb9fb8ff-kx2ps   1/1     Running   0           10m
currencyservice-68574d58b5-t52m2   1/1     Running   0           10m
emailservice-666fb7787c-wxvz8      1/1     Running   0           10m
frontend-74fbd86d4f-n947x         1/1     Running   0           10m
loadgenerator-76556ff55b-drrg4     1/1     Running   0           10m
paymentservice-5cf79b4f59-rmr4p    1/1     Running   0           10m
productcatalogservice-7dcdb9775c-rsdrw 1/1     Running   0           10m
recommendationservice-666b55c4d7-pbkz7 1/1     Running   0           10m
redis-cart-5cc6c9f86c-gkl2z        1/1     Running   0           10m
shippingservice-77dd667c99-x8h88    1/1     Running   0           10m
[root@ip-172-31-30-22 ~]#
```

